

# Ayush Gupta

+91 9702402372

ayushg1500@gmail.com

ayush-gupta

ayush-gupta

## EDUCATION

### Electronics and Telecommunication Engineering

Sardar Patel Institute Of Technology

Nov. 2022 – Present

Mumbai, India

### HSC

Nirmala Memorial Foundation College of Science

Aug. 2021

Mumbai, India

### CBSE

MKVV International Vidyalaya

May 2019

Mumbai, India

## WORK EXPERIENCE

### Software Development Intern

SP-TBI

Feb. 2025 – May 2025

Mumbai, India

- Built a **ticket-based room booking system** using Python, Django, and PostgreSQL, replacing manual approvals and reducing request processing time from approximately 10–15 minutes to under a minute.
- Integrated 5+ backend modules, including automated email notifications and role-based authentication, enhancing system responsiveness and security, and earning positive feedback from internal testers on usability and efficiency.

## PROJECTS

### Jasdaq: Low Latency Stock Trading Engine | Java, Spring Boot, MySQL

Aug. 2025

- Built a low-latency stock trading engine in Java with a price–time priority limit order book using **self-implemented data structures** (Red–Black Trees) for  $O(\log n)$  lookups and  $O(1)$  order management.
- Eliminated standard-library overhead, reducing **GC pressure** and **runtime allocations**, and processing 100,000 orders in **12 ms** (120 ns/op).
- Designed a **single-threaded matching engine** with **multi-threaded input** handling via a **BlockingQueue**, achieving **thread-safety**, **low lock contention**, and **deterministic sequential execution**.
- Delivered low-latency market updates through lightweight Spring Boot **WebSocket** interface for interactive, real-time clients, while leveraging **UDP multicast** and **Kafka** for ultra-fast, one-to-many trade notifications.

### EduMerge: Content Aggregator | Django, React, MongoDB, MySQL

June 2025

- Devised a full-stack course aggregator with **JWT-authenticated** and **OAuth-enabled role-based access control (3 distinct roles)** using **Django, React**, and **Tailwind**.
- Automated data ingestion of **1000+ courses** from **MIT OCW** and the **YouTube API** via a **Selenium-based** scraping pipeline backed by **MongoDB**, achieving **90%+ sync accuracy** in under **2 minutes**.
- Enabled user uploads with **AWS S3** and **Cloudflare Stream**, delivering **100+ videos** via structured **MySQL** metadata.

### Glimpse: Blog Website | Flask

Nov. 2024

- Developed **Glimpse**, a responsive blog platform using **Flask** and **Bootstrap**.
- Designed a secure admin panel with **RBAC** and full **CRUD** capabilities, reducing management time by **30%**.
- Implemented blog **search functionality** ( $O(n)$  search, tested with **200+ posts**) and a **functional** contact form integrated with **Flask-Mail** for simulated email delivery.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, SQL

**Tools & Platforms:** Linux, Git & GitHub, Postman

**Frameworks & Libraries:** Django, Flask, React, DRF

**Databases:** MySQL, MongoDB

**Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks

## POSITION OF RESPONSIBILITY

### Training and Placement Coordinator | S.P.I.T

June 2024

- Coordinated end-to-end campus recruitment processes for firms like WorkIndia, Barclays, and Nomura, acting as the liaison between students and recruiters.
- Maintained and optimized placement databases for current and graduated students, enabling real-time tracking and data-driven decision making.